Available online at www.sciencedirect.com

ScienceDirect

Procedia Economics and Finance 8 (2014) 484 – 489

Procedia Economics and Finance

www.elsevier.com/locate/procedia

1st International Conference 'Economic Scientific Research - Theoretical, Empirical and Practical Approaches', ESPERA 2013

How rural agricultural development projects (animal production) can use projects benefits for improving the economics states of farmers

Mostafa Abdel Rahman Ibrahim[a], Mocuta Dorina[b]*, Ibrahim Abdelrazek[c]

[a]*University of Kafr El Sheikh,Egypt*
[b]*University of Agricultural Sciences and Veterinary Medicine, Bucharest,B-dul Marasti, No.59, 011464, Bucahrest, Romania*
[c] *University of Kafr El Sheikh, Egypt*

Abstract

These projects are aimed especially for developing countries, where people focus on improving health and giving them the chance to rise from hunger and poverty.Each participant in such a project has multiple objectives. A farmer is interested in ensuring the welfare of his family, to educate children, to increase their income. Therefore, we will try to improve, for example, animal breeds, having a business development plan that take into account potential risks.
As a major objective for any private companies or government corporations, the maximization of net income and gross added value, holds an important place for improving the corporate financial behaviour. On the other hand, making the highest profit possible in any activity implies reducing the risk. By applying policies to support farmers, society as a whole, may have important benefits as such: increase the number of jobs and the unemployment rate drops. Such projects represent an important investment in rural development, which provide good income for future with low costs and great benefits.

© 2014 The Authors. Published by Elsevier B.V. Open access under CC BY-NC-ND license.
Selection and peer-review under responsibility of the Organizing Committee of ESPERA 2013

*Keywords:*rural development, benefit farmers, risks, reproduction sheep;

* Corresponding author. Tel.:+4-072-183-5835.
*E-mail address:*dorinamocuta@yahoo.com

2212-5671 © 2014 The Authors. Published by Elsevier B.V. Open access under CC BY-NC-ND license.
Selection and peer-review under responsibility of the Organizing Committee of ESPERA 2013
doi:10.1016/S2212-5671(14)00117-8

1. Introduction

Rural development is a topical issue, it wanting to achieve a balance between maintaining rural values and the trend of modernization of rural life on the other side. As defined by the European Union, we can say that: "Development Rural is a means to improve the overall economic and social welfare, the rural residents and the institutional environment in which they live ".

From this acceptable rural development policy objectives of the European Union, as defined in Regulation Council no. 1257/17 May 1999 on support for rural development covers:

- The improvement in agricultural holdings;
- Ensuring the safety and quality of agricultural products;
- Ensure fair and stable levels of farmers' income;
- Environmental Protection;
- Development of complementary and alternative activities generating jobs work to counter the depopulation of agricultural areas and strengthen substance rural economic and social;
- Improving working and living conditions in rural areas and promoting equal opportunities.

1. Livestock is a key of improving income

Livestock is a major factor in increasing the general contribution to household livelihoods level by providing a variety of direct and indirect pathways for agricultural development. As it is presented in literature (Alary et al., 2011; Moll et al., 2007), livestock provide cash income or income in kind through the sale of animals and/ or the sale and consumption of milk, meat, eggs and other animal products.

On the other hand, livestock are an economic (capital growth by increasing herd) and insurance for the sale of animals is get immediate revenue, which helps cover some important and unexpected expenses (eg. school or charges care) .

Third, animals produce manure can be used in transport services, both in our own farm , but they can be rented or sold for the same purpose (for example, hiring a bull for plowing) . Also a source of wealth, animals help maintain a certain social status , enabling access to financial services .

Also, some animals may be kept close to home (for example poultry)labor required is small. Thus, especially women, can manage both these activities, and others (eg. cooking or childcare) which requires time and are able to provide the necessary food for the family and even the market.

As (Alary et al., 2011; Moll et al., 2007) notice, it is difficult to quantify the overall contribution of livestock to household livelihoods, and only few have tried. The International Funds for Agricultural Development and its partners in developing countries share the challenging mission of enabling the rural poor to overcome their poverty. This partnership materialises primarily through hundreds of rural development projects and programmers at grass-root level, poor animals, poor food and low animal protein level, often in the poorest and most remote regions of the world. Although the performance of these projects has improved in many aspects over the near years, external and internal evaluations continue to report weaknesses in their monitoring and evaluation, in particular in the way impact of the benefit and results which are carried out and used at project management and field application.

The livestock sector contributes over 21% of agricultural GDP (6% of national GDP), of which pig production accounts for 71% of livestock output. The recent increase in livestock production has been driven by rising domestic demand, particularly in urban areas where per capita incomes have risen fastest. In general, meat demand has been rising among a growing population of consumers that is increasingly urbanized. As a lucrative example (Heffeman, 2004) brings into discussion the situation of the poor farmers in Mali for who the small ruminants continues to remain the main source of livelihoods.

(Heffeman, 2004) argues that, “sheep and goats assets are key opportunities for smallholder small ruminant producers to not only engage in income generating activities enabling them to escape the poverty trap but also to consume animal source food they could not afford to buy.”

In the other study (ILRI, 2011) it is reviled the importance of the aquaculture sector in providing high degree of income contribution, employment creation and food security in Egypt, all of which are national priority areas given low per capita income levels, rising population, worsening food security indicators, and official unemployment

levels which have remained at around 10% for the last ten years. (ILRI, 2011).

There was widespread concern that the knowledge base to address the pressing problems facing agriculture was not being generated quickly enough. Many felt that the problems will get progressively more challenging as population growth places more pressure on agricultural productivity and on the natural resource base.

Around the world, small pet owners are almost 20 % of world population and occupy a large part of the agricultural land in the tropics. In the future, expect an increase in demand for animal products, especially in developing countries, which provide an important opportunity to increase living standards, also improving environmental management. This has to be supported by policies that allow investments in support of the poor. In enhancing support systems for small animal owners, we must consider the social welfare, which is directed in areas and sectors with best social, natural resources provided to allow this.

Small pet owners, can straighten, especially in ruminant systems , particularly dairy products, where you can use family labor and the ability of ruminants to use a low quality roughage . Small pet owners are pawns basic local markets , especially important in developing countries agriculture .

Therefore, as the production and marketing systems evolve , they will have little help pet owners by providing effective services, links to markets and production will be taken to mitigate risks. Also need support from the public sector for the poor to adapt and benefit from evolving systems. Is need also to be intensified livestock production sectors and source of income. Areas surrounding watershed will require less investment. They need to move especially in large areas of Africa, fuelled only by rain. Therefore, intervention policies to support adequate investment are needed to avoid irreversible damage to ecosystems.

Attention should be directed toward social and environmental sustainability, so animal production, to be stimulated in poor countries , thus reduce the gap between rich countries and poor countries . We give two examples , in this sense , sustainable intensification : can be illustrated , smallholder dairy systems in East Africa and South Asia and small ruminant meat systems in sub- Saharan Africa.

In a free market economy smallholders are playing complex and different roles in order to achieve the best results and also have a high degree rate of dependents of their inlands economies. In this context the World Development Forum provides important data on key livestock drivers, taking into account the fact that livestock makes an important contribution in economic value and in particularly in agriculture-based economies as Egyptian economy is.

Countries as China, Vietnam and India has highest and most rapid increasing rates in GDP growth which has imposed a big demand for livestock products especially during 1990 to 2005. The livestock provides great revenues and large portion of GDP contribution in these countries, especially “in societies as daily per capita income moves from US$ 2 to US$ 10”. (ILRI, 2006).

As (Herrero et al., 2009a) remarks “at income levels of US$2 or less a day, households are pre-occupied with meeting their basic need for calories from the cheapest source. Under such poverty levels, food expenditure are made on relatively cheap sources of calories, such as cereals and roots and tubers, and the consumption of livestock products is low”. (Herrero et al., 2009a).

For households with an income of U.S. $ 2 per day, the request for animal products increases, especially in urban areas .Those whose income will increase, will become consumers for high quality animal products. In the situation where per capita daily income is below $ 5 overrides quantity at the expense of , quality . These are key elements of the application, producing changes structural livestock production, processing and distribution. A large part of the small producers, selling their livestock products, low income consumers, guided by informal markets. Although urbanization and incomes have increased, however, they depend on the length and complexity of livestock value chains, the quality and safety standards required in animal markets, it is difficult for small farmers, thus forced to compete these growth markets. It is thus necessary agreements between primary producers, processors and distributors to meet these conditionssophisticated .

These structural changes in livestock value chains typically involve new actors such as private agribusiness firms thus provoking organizational and institutional innovations. Pro-active policies and investments can help ensure the inclusion rather than the exclusion of the poor who produce and sell on a small scale (Kaitibie et al., 2008).

A household survey, conducted in 12 developing countries shows that livestock are a common asset for rural households in all expenditure quintiles. A percentage of 68 per cent of rural households in the lowest two quintiles of expenditure kept some farm animals , and the percentage drops to 58 per cent for households with a higher

income, higher echelon . Most of the households, for some reason, do not give animals as either a ISK - coping strategies or increase revenue or be due to lack of means, or have fewer opportunities to diversify into non - livestock income activities is - strengthening. There is a balance of livestock (TLU) distributed among the population, keeping animals, except for Latin American countries , the composition of the marginal effect size correlated with household wealth. Poor households prefer to keep small ruminants compared to the richest, where large ruminants are preferred. In countries used as the sample, the probability of keeping, large or small ruminants varies with the scale of 1 to 96 per cent , and property do not remain the same pattern in the sample countries. Chicken owners are spread equally in all quintiles of expenditure in all countries except Nepal (53%). Therefore, we consider that the policies and investments, which refers to small ruminants and poultry directly benefit, a higher proportion of poor households keeping animals compared with interventions related to large ruminants.

The new technology is the basis of agricultural productivity growth, which must be scientific and technological skills, a system of permanent renewal of technology and other services to farmers, as farm management and commercial orientation. In particular, in areas of Asia, small farmers showed great ability in using the new technology when they received incentives, financial support and appropriate infrastructure. Also, small farmer's cash - African cultures showed small potential on the continent, and in recent years have had achievements in food crops. Deprived areas of ecological, rural population, poor land owners have not benefited from advances in technology and do not know the progress, if governments are not willing and not able to redistribute land and resources, providing support and incentives necessary.

2. Egypt as a study case.

2.1. Research in Egypt goals, accomplishments and needed improvements

With a population now 81 million people, Egypt has a growing population will exceed 140 million by 2050. A threat to food security is the limited water resources (actual water resources per capita are 900m3) , and the resources of arable land (only 3.5 per cent of total land area). Also, the land resources are negatively impacted by a number of factors desertification. A number of challenges for food production and food security is the climatic conditions on the growth temperature and high evapotranspiration.

Egypt is a country wide, covering an area of about one million km2 , being adjacent to the Mediterranean Sea to the north , east of the Red Sea (coastal Total 3,500 km), and western Libya.

In general, the climate is dry, hot desert. In Lower Egypt during the winter, the climate is mild, sometimes with rain in coastal areas and in Upper Egypt is rainless with sunny days, warm and cold nights. The climate is hot and dry in summer, throughout all Egypt.

Much of Egypt is desert land. Arable land is 3.5 per cent of arable land and it is about 8 million acres of "old" , located in the Nile Valley , and 2 million acres , is recovered ground . Extending through land reclamation cannot be achieved because of water scarcity and misuse of water. Also, Egypt's agricultural production has taken into consideration the fact that the population in the country is growing.

Achieve the last two decades are numerous, have developed new varieties, agronomic practices were applied improved livestock developed and have been used better techniques for food processing.

In recent decades, Egypt has gained a better understanding of soil and water systems, but implement this new knowledge was performed in small measure, due to the fundamental management issues. Collective effort is required to maintain a sound scientific basis, applied science and a strong extension service.

2.2. Egyptian research for agriculture today

Although it has limited natural resources and a rising populatio, Egypt is an intensive user of modern technology, aimed at improving agricultural productivity. In this sense the early 20th century, the Ministry of Agriculture (MALR) with technical divisions , enhanced research capabilities .

Thus was created the Agricultural Research Center (ARC) in the early 1970. In the last two decades, noting many achievements, including the development of new varieties, improved agronomic application of methods , animal development , maintenance of national herds and food processing by technologies. Along with basic science have

introduced new crops and breeds of animals and research has been devoted to solving problems. The primary objective was to maximize the economic return per unit of land and water. Agricultural Research Center (ARC) implemented so far, five 5-year plans. ARC has the following major functions in agricultural development strategies at national level:

• to generate a continuous flow of technology to increase productivity and reduce production costs, it is necessary to conduct applied and basic research ;

• transfer of new technologies in the agricultural community to achieve with extension service and monitoring and adoption by end users also

• Human capital development will be an on-going process. ARC as its founding law will track infrastructure development, establish priorities, to pursue training or research staff, and support staff and upgrade its physical capacities in order to achieve, a greater durability. In twenty years, ARC research staff increased from 1,720 researchers in 1982 to 4300 researchers in 2001. Central laboratories and institutes we were added to contribute to a better performance.

On going plan is built on the following topics pivot:

• sustainable research and extension research capabilities ;

• updating technology transfer channels.

It is necessary to use at a maximum level, the findings of science and modern technology applied abroad. The fifth 5-year plan, has 14 research programs, is included in 16 institutes, 13 central laboratories, 10 regional stations,36 stations specific research, 21 research administration throughout Egypt and four research extension and training centres of excellence. MALR partner agencies, the Ministry of Water Resources and Irrigation (MWRI), universities and research centres sisters support this effort.

The current plan, apply a multidisciplinary approach which is a major feature and key to its success. It requires a clear definition of research topics that solve specific problems, complete with objectives to be achieved, and to establish, in accordance with natural resources, human and financial resources allocated.

Important topics for which efforts are made in Egypt are : increasing food production and employment , the management of natural resources, especially water and fertilizers, proper selection of crops and varieties , increased attention to improve animal production and adaptation of products for export but also for domestic markets. Integrated research requires institutional mechanisms, collaborative strategy and structures capable, to unite their efforts in and interests, research organizations and customers, so as to enhance interactions among the various components of research.

3. Romania in socio-economic context of rural

Romania has significant growth potential, although insufficiently exploited. With a total area of 238,000 km and a population of over 21 million people, Romania is the largest, the second new member state of the European Union, after Poland. Rural areas have a substantial growth potential and a vital social role. According to the definition 1 of the National Rural Areas Legislation in Romania covering 87.1% of the country, Comprising 44.9% of the Population (National Statistics Institute) namely 9.63 million people in 2010 compared to 9.65 million in 2009 (INS 2011).

Due to these factors, Romania has signed funding agreements with the International Monetary Fund and the European Union. These agreements ensure adequate financing current account deficit and send a more credible economic and financial policies. Human capital presents particular importance for rural development. Rural development and diversification of the rural economy depends on the level of education, knowledge and skills. Romania will have to follow a husbandry promote the multifunctional role and influence in the countryside. To this should be considered animal husbandry practice, taking into account the preservation of traditions, maintaining specific areas, the maintenance activities of rural communities, maintaining rural population, especially young people.

Development of livestock farms are a great means of transformation products superior biological properties of various plant resources and food industry by-products can not be used directly for human consumption. Regarding the sheep / goats, they increased in the period 2005-2010 (except in 2006 when there was decrease) from 8.921

million head (2005) to 11.177 million head in 2010. Sheep are only species of the animals which Romania has exported more than imported. (INS, 2011).

Conclusions

- Agriculture plays an important role strategically in all countries. It is the main sector that ensures food security of the population;
- Agriculture contributes to sustainable economic development and environmental protection;
- Rural development policy can ensure stable and fair income for farmers;
- Farms are generating jobs improving the living standard of the inhabitants.
- Acquisition of Structural Funds is a goal meant to bring a major contribution, rural development, encouraging the establishment of farms, with benefits to improve product quality, resulting from the application of modern technologies.

References

Councilofthe European Union, COUNCIL REGULATION (EC) No 1255/of 17 May 1999.

Alary, V., Corniaux, C. and Gautier, D. (2011) Livestock's Contribution to Poverty Alleviation: How to Measure It? ,World Development forthcoming.

ILRI, 2011, Pig production benefits rural livelihoods in Vietnam.

H. Steinfeld and S. Mack,2012. Livestock development strategies, Sustenable Rural Development Trough Agriculture.

Heffernan C.2004. Livestock and the Poor: Issues in poverty-focused livestock development1 Issues in poverty-focused livestock development. Chapter 15, in: Responding to the Livestock Revolution: the role of globalisation and implications for poverty alleviation. Eds. E. Owen, T. Smith, M.A. Steele, S. Anderson, A.J. Duncan, M. Herrero, J.D. Leaver, C.K. Reynolds, J.I. Richards, J.C. Ku-Vera. British Society of Animal Science, publication 33.

Oakley, R., 1998. Experiences with community-based livestock workers programmes, methodologies and impact: A literature review. The Veterinary Epidemiology and Economics Research Unit, University of Reading, Reading, UK.

Martin, M., 2001. The impact of community animal health services on farmers in low-income countries: A literature review. VETAID, UK.

Moll, H.A.J., Staal, S. and Ibrahim, M.N.M. (2007) Smallholder dairyproduction and markets: A comparison of production systems in Zambia, Kenya and Sri.

Lanka. Agricultural Systems 94(2), pp. 593-603.

Heffernan, C., Misturelli, F. and Nielsen, L., 2001. Restocking and poverty alleviation: Perceptions and realities of livestock keeping among poor pastoralists in Kenya. Report for DFID's Livestock Production Programme, NRI, Greenwich;

J.J. McDermott, S.J. Staal, H.A. Freeman, M. Herrero, J.A. Van de Steeg,2010, Sustaining intensification of small holder livestock systems in the tropics, Livestock Science xxx (2010).

ILRI, 2006. Lessons from a changing world: implications for livestock research and development. In: Birthal P.S., Taneja V.K. and Thorpe W. (eds). Smallholder Livestock Production in India: Opportunities and Challenges. Proceedings of an ICAR–ILRI international workshop held at National Agricultural Science Complex, DPS Marg, Pusa, New Delhi 110012, India, 31 January–1 February 2006. NCAP (National Centre for.Agricultural Economics and Policy Research)–ICAR (Indian Council of Agricultural Research), New Delhi, India, and ILRI (International Livestock Research Institute), Nairobi, Kenya.pp 65–112.

Herrero,M., Thornton, P.K., Notenbaert, A.,Msangi, S., Wood, S., Kruska, R., Dixon.J., Bossio, D., van de Steeg, J.A., Freeman, H.A., Li, X., ParthasarathyRao, P., 2009a. Drivers of change in crop–livestock systems and their potential, impacts on agroecosystems services and human well-being to 2030.CGIAR Systemwide Livestock Programme.International Livestock Research Institute, PO Box 30709, Nairobi00100, KenyaKaitibie, S., Omore, A., Rich, K., Salasya, B., Hooton, N., Mwero, D., Kristjanson, P., 2008. Influence pathways and economic impacts of policy change in the Kenyan dairy sector: the role of smallholder dairy project. Research report for the CGIAR Standing Panel on Impact Assessment, Nairobi, p. 64.

Lowell N. L.,2008. EGYPT'S FUTURE DEPENDS ON AGRICULTURE AND WISDOM , World Opinion Recognizes Agriculture World Development Report In announcing "Agriculture for Development" as the theme Livestock Assets, Livestock Income and Rural Households Cross-Country Evidence from Household S Ugo Pica-Ciamarra, Luca Tasciotti , FAO, Rome, ISS, The Hague; WB, Washington D.C., May 2011.

Joint paper of the World Bank, FAO, AU LivestockAssets,LivestoK Income and Rural Householdscountry Evidence from Household Surveys, Tasciotti L., Alberto J., Zezza.2008. WB, Washington D.C.the World Bank, FAO, AU-IBAR, ILRI with support from the Gates Foundationof the World Development Report (WDR) for 2008, (127).

National Statistics Institute, INS 2011.